Supplementary material: Fiducial Pk-Ratio Prediction on the Full Emulator k-Grid

HEP-Genesis-Agent session record
Project: expert-schneider15-cl-bias

2026-10-25

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `20adb112`
- Started: 2026-10-25 16:24:05; report generated: 2026-10-25 16:40:48
- Project: expert-schneider15-cl-bias (`/Users/nramachandra/hep-genesis-testing/projects/expert-schneider15-cl-bias`)
- Compute execution site(s) selected in the session: local
- 2 user turn(s), 3 tool call(s) (0 returned errors), 0 figure(s), 1 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`gpt-5.6-luna@openai`	`https://mag.genesis.americansciencecloud.org/v1`
report	`gpt-5.6-luna@openai`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`gpt-5.6-luna@openai`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`

Server	Transport	Location
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`serve-hydro.sh describe_parameters`	`{"stat_name": "Pk-ratio"}`	`{ "status": "success", "message": "7-parameter design space for Pk-ratio: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { "ka...`
2	`serve-hydro.sh describe_parameters`	`{}`	`{ "status": "success", "message": "7-parameter design space for all statistics: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { ...`
3	`serve-hydro.sh predict_observable`	`{"stat_name": "Pk-ratio", "omega_m": 0.143, "sigma_8": 0.82, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "z": 0.5, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/expert-schne...`	`{ "status": "success", "message": "Predicted Pk-ratio at z=0.5000 (799 bins).", "files": [ "/Users/nramachandra/hep-genesis-testing/projects/expert-schneider15-cl-bias/runs/261025_162428/Pk-...`

A.5 Files produced

- `Pk-ratio_z0.5000_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testing/projects/expert-schneider15-cl-bias/runs/261025_162428/Pk-ratio_z0.5000_prediction.csv`